

Priyanshi Nigam

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SUMMARY

Computer Science undergraduate at VIT Bhopal University and an aspiring **Machine Learning Engineer** with hands-on experience building and deploying ML/DL models in Python. Good problem-solving and analytical skills with the ability to develop practical technology-driven solutions.

EXPERIENCE

Machine Learning Intern – RSMML

May 13 – Jun 30, 2026

- Built an end-to-end ML system at **Rajasthan State Mines and Minerals Limited (RSMML)** predicting ore quality (% Silica Concentrate) from 737K+ flotation sensor records across 22 process parameters
- Performed EDA (correlation heatmaps, distribution analysis) and feature scaling; trained & benchmarked 4 models (Linear Regression, Decision Tree, Random Forest, ANN), with Decision Tree achieving **R² of 0.99** and ANN reaching **R² of 0.94**
- Applied a 70:30 train-test split and cross-model evaluation (R², RMSE, MAE); presented findings to IT leadership for real-time process control

PROJECTS

Brain Tumor Detection & Classification (EfficientNetB0)

[GitHub](#) | [Live Demo](#)

- Re-engineered a 4-class MRI tumor classifier by fine-tuning EfficientNetB0 with a custom dense head (BatchNorm + progressive Dropout), lifting accuracy from 82% (VGG16) to 93.1% on 1,324 scans
- Handled class imbalance via computed class weights, applied heavy augmentation (rotation, shift, flip, zoom, brightness) at 224×224, and used early stopping with LR scheduling
- Achieved weighted F1-score of 0.93 and per-class precision of 87–98%; deployed as an interactive Streamlit web app for real-time tumor prediction

Paraphrase Generation using T5 Transformer

[GitHub](#) | [Live Demo](#)

- Fine-tuned a T5-base Transformer on the PAWS dataset (6,000 pairs) for paraphrase generation, achieving BLEU 63.90 and ROUGE-L 0.845
- Optimized training with FP16 mixed precision and gradient accumulation; built a full tokenization, training, and beam-search/sampling inference pipeline
- Deployed the fine-tuned model to Hugging Face Hub for public, API-based paraphrase generation

Drug Discovery Pipeline for EGFR Inhibitors using ML

[GitHub](#)

- Collaborated on a cheminformatics pipeline sourcing EGFR bioactivity data (ChEMBL API) and molecular fingerprints (MACCS, Morgan)
- Built and trained an ANN (Keras) to predict pIC50 bioactivity, tuning hidden-layer size and batch size
- Evaluated performance (R², MAE, MSE) and applied the model to rank top drug candidates

EDUCATION

B.Tech in Computer Science (AI & ML) VIT Bhopal University

2023-2027 – *CGPA: 8.92/10*

Class XII, CBSE Board

2023 — *86.6%*

Class X, CBSE Board

2021 — *94%*

SKILLS

Domain	AI/ML, ML Engineering, Deep Learning, NLP, LLM, Computer Vision, Model Deployment
Languages	Python, C++, SQL, Java(basics)
Machine Learning & Deep Learning	TensorFlow, Keras, ANN, CNN, Transfer Learning, RAG, VGG16, EfficientNetB0, T5
Tools & Platforms	Git, GitHub, Hugging Face, Streamlit, REST APIs, Jupyter, Google Colab, HTML, CSS
Data Analysis	NumPy, Pandas, Matplotlib, EDA, Data Visualization
Professional Skills	Problem Solving, Teamwork, Communication, Fast Learner, Project Management

CERTIFICATIONS

Applied Machine Learning in Python

Coursera

Microsoft Azure Data Fundamentals (DP-900)

Microsoft